Research Paper Extraction

1. **A Survey on Hybrid-CNN and LLMs for Intrusion Detection Systems**

- **Objective:** To critically evaluate AI-driven Intrusion Detection Systems (IDS) for IoT using the most recent datasets.
- **Methodology:** Comparative benchmarking of deep learning and Large Language Models (LLMs) on datasets like Ton_IoT, CICIoT2023, and X-IIoTID.
- **Algorithms/Models:** CNN-LSTM, BERT, RNN, and GRU.
- **Key Results:** BERT achieves exceptional validation accuracy (99.99%) but suffers from high inference time (688 seconds on large datasets). Hybrid models like CNN-RNN balance accuracy and speed.
- **Limitations:** LLMs and large hybrid models require significant computational resources, limiting real-time deployment on edge devices.

2. **AI-Powered CCTV Surveillance with YOLOv5 and Raspberry Pi**

- **Objective:** To create a cost-effective, real-time surveillance system for detecting unauthorized humans or vehicles.
- **Methodology:** Deploying a YOLOv5 model on a Raspberry Pi 4B for live camera feed processing.
- **Algorithms/Models:** YOLOv5n (Nano version).
- **Key Results:** Achieves ~85% daytime accuracy, processing speed of ~5 FPS, and alert delivery under 3 seconds.
- **Limitations:** Slower FPS on Raspberry Pi, poor accuracy in low-light conditions without IR, and struggles with high-speed objects.

3. **Leveraging AI for Intrusion Detection in IoT Ecosystems: A Comprehensive Study**

- **Objective:** To analyze the performance and scalability of AI methodologies for identifying diverse cyber threats in IoT.
- **Methodology:** Comprehensive literature review categorizing ML/DL techniques and their mathematical foundations.
- **Algorithms/Models:** SVM, Decision Trees, Random Forest, CNN, RNN, and Reinforcement Learning.
- **Key Results:** Deep learning architectures consistently outperform traditional ML; models by Rahman M.A. and others exceed 99% accuracy.

- **Limitations:** Resource constraints of IoT devices and lack of high-quality datasets for Advanced Persistent Threats (APTs).

4. **Private Blockchain Envisioned Access Control System**

- **Objective:** To design a secure access control mechanism for Industrial IoT (IIoT) edge computing using blockchain.
- **Methodology:** Simulation study using the MIRACL library to measure computational time on servers and Raspberry Pi 3.
- **Algorithms/Models:** Elliptic Curve Cryptography (ECC) and SHA-256 hashing.
- **Key Results:** Demonstrates lower communication and computation costs than existing certificateless schemes.
- **Limitations:** Total computational time increases exponentially with the number of transactions and P2P nodes.

5. **Scalable Intrusion Detection via Property Testing and Federated Edge AI**

- **Objective:** To reduce computational overhead in IoT IDS through sublinear-time sampling.
- **Methodology:** Integrating property testing with Federated Learning and Edge AI for decentralized monitoring.
- **Algorithms/Models:** Lightweight CNN and Chi-Square statistical property tests.
- **Key Results:** 70% reduction in processing time and 92% accuracy while analyzing only 10% of the data.
- **Limitations:** Lacks validation in live, unpredictable IoT environments with hardware heterogeneity.

6. **Semi-Supervised Federated Learning (SSFL) Based IDS**

- **Objective:** To address privacy leakage and high communication overhead in distillation-based federated learning.
- **Methodology:** Developing a semi-supervised FL scheme using unlabeled data and a knowledge distillation voting mechanism.
- **Algorithms/Models:** CNN-based classifier and discriminator networks.
- **Key Results:** Higher detection performance (87.4%) and significantly lower communication overhead than traditional FL.
- **Limitations:** High dependence on the availability of an initial unlabeled open dataset.

7. **Hybrid IoT Security Architecture: ML & Blockchain IDS**

- **Objective:** To enhance IoT security through optimized feature selection and immutable logging.
- **Methodology:** Using nature-inspired optimization and attention-based deep learning.
- **Algorithms/Models:** Red Fox Optimization (RFO) and Attention-based BiLSTM.
- **Key Results:** Achieves ~98.9% accuracy with improved scalability via blockchain-based secure logging.
- **Limitations:** Significant blockchain computational overhead and resource constraints on edge devices.

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2. Comparison Table

Technique	Performance	Advantages	Drawbacks
YOLOv5 Object Detection	~85% Accuracy, 5 FPS	Real-time, low-cost (<₹5000)	Low FPS on Pi, fails in low light
LLMs (BERT)	~99.99% Accuracy	Contextual understanding, high precision	Extremely high latency/testing time
Federated Learning	84-87% Accuracy	Data privacy, decentralized training	Vulnerable to model poisoning/drift
Property Testing	92% Accuracy @ 10% Sample	70% reduction in processing time	Probabilistic; may produce false positives
BiLSTM + Blockchain	~98.9% Accuracy	Immutable logs, high feature precision	High computational overhead
Hybrid CNN-LSTM	~93-99% Accuracy	Captures spatial-temporal features	Longer training times than pure RNNs

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3. Patterns

- **Frequent Methods: Hybrid CNN-based architectures** (CNN-LSTM, CNN-RNN) are the most frequent choice for capturing both spatial and temporal data patterns. **Federated Learning** is the standard approach for privacy-preserving distributed IoT systems.

- **Best Results: Fine-tuned BERT** models and **Attention-based BiLSTM** models yield the highest accuracies (near 100%), though they often require GPU acceleration or cloud offloading for practical speeds.

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4. Research Gaps

- **Resource Constraints vs. Complexity:** High-performance models (LLMs, BiLSTMs) are often too heavy for ultra-low-power microcontrollers like ESP32/STM32.
- **Real-World Volatility:** Most studies rely on benchmark datasets (NSL-KDD, CICIDS2017) that do not capture the unpredictable noise and dynamic topologies of live production environments.
- **Adversarial Robustness:** While some models mention FGSM/PGD resilience, many lightweight IoT systems remain vulnerable to crafted evasion attacks.

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5. Suggested Improvements for Your Project

- **Project Context:** AI-Powered CCTV with YOLOv5 and Raspberry Pi.
- **Pruning and Quantization:** Apply **TensorFlow Lite quantization** (model compression) to increase YOLOv5 FPS on the Raspberry Pi from 5 to near 15-20 FPS.
- **Multimodal Detection:** Integrate a **network-based IDS agent** alongside the visual YOLOv5 model to detect if a "hacked" CCTV camera is performing lateral movement or data exfiltration.
- **Secure Alert Logging:** Add a **Private Blockchain layer** to ensure that once an intruder is detected, the snapshot/log becomes immutable and cannot be deleted by a compromised admin account.
- **Active Night Vision:** Use **IR cameras** to overcome the limitation of poor low-light accuracy mentioned in the project documentation.

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6. Ready-to-Use Content

Literature Survey Paragraph

Recent advancements in Internet of Things (IoT) security emphasize the transition from passive monitoring to active, AI-driven Intrusion Detection Systems (IDS). Surveys of state-of-the-art methodologies reveal that while monolithic Machine Learning models are being phased out, hybrid architectures combining Convolutional Neural Networks (CNN) for spatial feature extraction and Recurrent Neural Networks (RNN/LSTM) for temporal analysis consistently achieve accuracies exceeding 93%. Furthermore, the integration of Large Language Models (LLMs) like BERT has set new benchmarks in contextual log analysis, albeit at significant computational costs that often preclude edge deployment. To mitigate these resource constraints, emerging frameworks utilize property testing for sublinear sampling—reducing processing time by up to 70%—and federated learning to preserve data privacy across distributed nodes. However, the trade-off between high-fidelity detection and the severe energy limitations of edge hardware remains a primary area of ongoing research.

Bullet Points for Viva Revision

- **Hybrid Models:** Combining CNN with LSTM/RNN is superior for capturing spatial-temporal patterns in IoT traffic.
- **Federated Edge AI:** Essential for data privacy as it trains models locally on devices without sharing raw data.
- **YOLOv5 for Surveillance:** A lightweight "Nano" model can provide real-time object detection on edge devices like Raspberry Pi.
- **Sublinear Sampling:** Property testing allows for 92% detection accuracy by analyzing only 10% of the network packets.
- **Blockchain Integration:** Provides immutable, tamper-proof logging for security alerts in industrial and residential settings.

5 Important Keywords

1. **AIoT (Artificial Intelligence of Things)**
2. **Federated Learning**
3. **Property Testing**
4. **Knowledge Distillation**
5. **Edge Inference**